

Ethical and Trust Consideration in the Integration of Chat Generative Pre-Trained Transformer (ChatGPT, version 3–4) into medical chatbots

100701434

University of Derby, Derby DE22 3AW
100701434@unimail.derby.ac.uk

Abstract: The integration of Chat Generative Pre-Trained Transformer (ChatGPT) into healthcare settings presents significant opportunities for improving patient care and healthcare delivery. However, this integration raises critical governance, trust, and ethical issues that require careful consideration. This paper analyzes the regulatory landscape, including EU regulations like the General Data Protection Regulation (GDPR) and Artificial Intelligence Act (AIA), and discusses vital considerations such as transparency, algorithmic bias mitigation, data privacy, and legal responsibility. It emphasizes the importance of developing an ethical governance framework to guarantee the conscientious and ethical utilization of ChatGPT within the healthcare sector. By placing importance on transparency, accountability, fairness, privacy, and safety, healthcare organizations can leverage the transformative capabilities of ChatGPT while maintaining ethical standards and protecting patient well-being. Collaborative efforts among stakeholders are imperative for tackling these obstacles and unlocking the complete advantages of AI-powered healthcare solutions.

Keywords: Chat Generative Pre-Trained Transformer (ChatGPT), Healthcare, Ethics.

Contents

1 Introduction.....	3
2 Context.....	3
2.1 Background on Generative AI Applications.....	3
2.2 Overview of Chat Generative Pre-Trained Transformer (ChatGPT, versions 3–4)...	3
2.3 The Objective of Integrating ChatGPT into Medical Chatbots.....	4
2.4 Role of Generative AI Technology in Medical Chatbots.....	4
2.5 Business Objectives and Implications of Implementing ChatGPT.....	5
3 Identification of Key Governance, Trust, and Ethical Issues.....	5
3.1 Principles of EU AIA, AI Harms Directive, GDPR, and Data for Humanity.....	5
3.2 The Cs of Generative AI Framework.....	6
3.3 Critical Governance, Trust, and Ethical Issues in ChatGPT Integration in Medical Chatbots.....	6
3.4 Justification of Identified Issues.....	7
4 Governance and Compliance Strategy.....	8
4.1 Impact of EU Regulations on Ethical and Trust Issues.....	8
4.2 Critical Evaluation of Governance and Compliance Measures.....	9
4.3 Development of Ethical Governance Framework for ChatGPT in Healthcare.....	9
5 Conclusions.....	10
6 References.....	11

1 Introduction

The integration of advanced artificial intelligence (AI), such as Chat Generative Pre-Trained Transformer (ChatGPT), into healthcare systems, has signaled the onset of a new era of possibilities for improving patient care and healthcare delivery. However, alongside these advancements come significant governance, trust, and ethical considerations that must be carefully examined and addressed. This paper delves into the landscape of integrating ChatGPT into medical chatbots, focusing on regulatory frameworks, ethical dilemmas, and the relevance of openness, answerability, and data privacy in guaranteeing the responsible utilization of AI in healthcare. Through examining these key issues, we aim to offer intuition into navigating the intricacies of AI integration in healthcare and fostering a healthcare environment that prioritizes patient well-being, ethical standards, and regulatory compliance.

2 Contexts

2.1 Background on Generative AI Applications

Lately, there have been remarkable innovations in artificial intelligence (AI), precisely in natural language processing (NLP), which focuses on comprehending and generating human language. One notable application within NLP is generative AI, enabling AI systems to create original and coherent content across various genres. Generative AI holds significant potential for integration into medical chatbots, which offers opportunities to enhance patient engagement and provide personalized healthcare information (Brown et al., 2020). Generative AI-powered chatbots, such as ChatGPT, can participate in natural and immersive interactions with users, facilitating communication in healthcare settings. These chatbots can help patients by answering their questions, supplying information about medical situations and treatments, and presenting support for managing health-related challenges (Roller et al., 2020).

2.2 Overview of Chat Generative Pre-Trained Transformer (ChatGPT, versions 3–4)

Chat Generative Pre-Trained Transformer (ChatGPT) has generated substantial fascination in several professions due to its sophisticated language model competencies (Guo et al., 2023). Developed by OpenAI, ChatGPT initially focused on tasks like language translation and answering simple questions but evolved with the release of GPT-4, which enabled comprehension of images alongside text (Lee et al., 2023; Sanderson, 2023). ChatGPT's rapid responses and natural conversational flow make it versatile across various fields, including healthcare, retail organizations, and software development (Roberts et al., 2020). In healthcare, integrating ChatGPT offers various opportunities and critical challenges, potentially revolutionizing research, patient interactions, and data analysis, (Hügler, 2023; King, 2023; Loh, 2023). From version 3 to 4, ChatGPT's evolution represents a significant advancement in AI, promising transformative applications, particularly in natural language processing and healthcare.

2.3 The Objective of Integrating ChatGPT into Medical Chatbots

The integration of ChatGPT into medical chatbots aims to achieve the following objectives:

- *Reducing Documentation Burden with ChatGPT:* One objective of integrating ChatGPT into medical chatbots is to alleviate the workload of clinical documentation, thereby reducing healthcare costs and mitigating professional burnout among physicians. Studies indicate that using ChatGPT significantly reduces the time and effort needed to write supervisory notes for complex cases (Barak-Corren et al., 2024). This demonstrates ChatGPT's potential to streamline documentation, allowing physicians to concentrate more on patient care and less on administrative tasks.
- *Enhancing CDS Alert Specificity with AI:* Integrating ChatGPT into medical chatbots is to enhance the specificity and effectiveness of clinical decision support (CDS) alerts. By generating unique and relevant suggestions for improving alert logic, ChatGPT can help optimize CDS systems. These AI-generated suggestions can play a crucial role in boosting CDS alerts, highlighting prospective upgrades to alert logic, and aiding in their execution (Liu et al., 2023). This exhibits the ability of ChatGPT to mitigate alert fatigue by refining the accuracy and relevance of CDS alerts, thereby supporting a more efficient and responsive healthcare environment.
- *Enhancing Medical Service Provision Through ChatGPT Integration:* The goal of incorporating ChatGPT into medical chatbots is to improve medical services by utilizing its capabilities to support medical laboratories, identify potential research topics, and keep healthcare professionals informed about the latest updates and advancements in various medical fields. ChatGPT is being utilized to assist medical laboratories, identify potential research topics, and provide the latest updates and developments to other healthcare professionals and students (Dave et al., 2023).

2.4 Role of Generative AI Technology in Medical Chatbots

Generative AI technology, as demonstrated by models like ChatGPT, has surfaced as a substantial transformative influence in the development of medical chatbots, revolutionizing various aspects of healthcare services and management (Cascella et al., 2023). By leveraging generative AI, medical chatbots can provide precise and personalized replies to user inquiries, thereby improving the quality of healthcare services. Through its skill to comprehend and produce natural language responses, ChatGPT facilitates effective communication between clinicians and patients, streamlining processes such as drug prescription, patient education, appointment scheduling, and medication management (Singhal et al., 2023). Moreover, generative AI-powered medical chatbots hold promise in being able to automate clinical documentation tasks by transcribing verbal interactions into structured electronic health records (EHRs), enhancing healthcare documentation's accuracy and efficiency (Harrer, 2023). Furthermore, the integration of generative AI in medical chatbots enables advanced functionalities such as real-time data retrieval, report generation, and decision support for clinicians, empowering healthcare providers with timely access to important information crucial for diagnostic and treatment decision-making processes (Sallam, 2023).

2.5 Business Objectives and Implications of Implementing ChatGPT

Incorporating ChatGPT into medical chatbots signifies a significant leap forward in healthcare management, offering a dynamic platform for seamless patient interaction and support, strengthened by advanced artificial intelligence capabilities (Mohd Javaid et al., 2023). By leveraging ChatGPT's capability to comprehend and respond to natural language inquiries like human conversation, healthcare organizations can ensure continuous and reliable assistance to patients, thereby promoting higher levels of satisfaction while optimizing operational costs through decreased reliance on human resources. Moreover, the integration of ChatGPT enables healthcare providers to streamline complex procedures like clinical decision support and automate tasks such as medical data interpretation and treatment recommendation generation (Mohd Javaid et al., 2023).

3 Identifications of Key Governance, Trust, And Ethical Issues

3.1 Principles of EU AIA, AI Harms Directive, GDPR, and Data for Humanity

- **Principles of the EU AI Act (AIA):** Under the projected EU AI Act, medical chatbots integrating large language models like ChatGPT would likely be classified as "high-risk" AI systems due to their potentially significant impact on human health and safety (Lan and Zhenjing 2022). As high-risk systems, they would be subject to stringent requirements such as human oversight to allow meaningful human control, transparency obligations to provide clear explanations for their outputs, technical robustness, and cybersecurity measures, with rigorous conformity assessments before being permitted for launch (Lan and Zhenjing 2022). Additionally, as a generative AI system, ChatGPT would have to adhere to specific transparency rules like clearly disclosing when content is AI-generated, implementing safeguards to thwart illegal content creation, and publishing synopses of any proprietary data utilized for training the used model (Winfield and Jirotko 2018).
- **AI Harms Directive:** The intended AI Harms Directive seeks to create a liability structure for AI systems within the EU. It suggests that operators of high-risk AI applications like ChatGPT-powered medical chatbots could be held strictly liable for any damage or harm caused by the AI system's outputs or advice (PMI 2022). Furthermore, the directive proposes a turnaround of the burden of proof in certain circumstances, making it easier for victims to claim compensation if they suffer harm due to inaccurate or harmful medical advice generated by AI (PMI 2022).
- **GDPR:** Using personal medical data to train large language models like ChatGPT raises significant secrecy worries under the EU's General Data Protection Regulation (GDPR). GDPR mandates that AI systems processing personal data must incorporate data protection rules from the initial design stage. (Winfield and Jirotko 2018). Additionally, the GDPR's purpose limitation and data minimization principles require that only the minimum mandatory individual information is gathered and processed for specified, legitimate objectives (Winfield and Jirotko 2018).

- **Data for Humanity:** There are risks of biased outputs if ChatGPT is trained on skewed medical datasets, necessitating ethical data governance principles from Data for Humanity, such as ensuring fair and responsible data use, data sovereignty to empower individuals over their data, and tackling data inequalities for equitable AI benefits (Müller 2012).

3.2 The Cs of Generative AI Framework

- **Control:** Control mechanisms are essential for ensuring that healthcare professionals maintain oversight and steer the behavior of Generative AI systems. This involves establishing governance frameworks that allow for the supervision and regulation of AI-generated outputs, such as those produced by patient education chatbots. By giving clinicians control over the content and ensuring that the technology complements rather than replaces human expertise, quality, and patient safety can be safeguarded.
- **Consequences:** Understanding the potential consequences of using Generative AI in healthcare is crucial for mitigating risks and ensuring patient safety. This includes expecting adoption barriers related to seeming value, usability, and risks, as well as defining interventions to drive acceptance. By promoting plainness, responsibility, and patient welfare, healthcare systems can harness the likely benefits of Generative AI technology whilst addressing systemic failures.
- **Correctness:** Correctness refers to the precision, reliability, and ethical soundness of AI-generated outputs in health care settings. Validated structures like the Technology Acceptance Model (TAM) and the Non-Adoption, Abandonment, and challenges to the Scale-Up, Spread, and Sustainability (NASSS) model offer practical strategies for implementing change management, investor consensus, and maximizing influence (Reddy, 2024). Ensuring that Generative AI systems produce factually accurate, evidence-based information is essential for building trust among healthcare professionals and patients. By upholding correctness through meticulous planning grounded in evidence and responsible governance, healthcare systems can realize the transformative potential of Generative AI while upholding their humanitarian obligations to provide quality, affordable, and humane care for everyone.

3.3 Critical Governance, Trust, and Ethical Issues in ChatGPT Integration in Medical Chatbots

- **Regulatory Compliance:** The integration of ChatGPT into medical chatbots necessitates robust governance frameworks to ensure compliance with regulatory standards such as HIPAA and GDPR (Ovalle et al., 2023). Preemptive and adaptable rules are required to guarantee that AI efficiently aids patients and mitigates harm (Zanca et al. 2022). Healthcare organizations must establish clear policies and measures for the ethical handling of AI technologies, which include data handling, secrecy protection, and accountability mechanisms. Even though OpenAI has employed secrecy measures and assurances to manage information in a masked form, the absence of adequate rules continues to be a fear (OpenAI (n.d.) 2023).

- *Transparency and Explainability:* These are crucial in algorithmic ethics. Transparency encompasses openness about how algorithms work, make decisions, and generate outputs (Balasubramaniam et al., 2023; Wachter et al., 2017). Explainability means providing clear reasons for an AI system's decisions (Williams et al., 2022). Both are essential for building trust among healthcare professionals and patients.
- *Algorithmic Bias:* Algorithmic bias, arising from design or decision-making processes within algorithms, occurs when they demonstrate biased behavior or produce unfair outcomes despite being trained on unbiased data (Kordzadeh et al., 2022; Akter et al., 2022). This bias stems from various sources such as biased feature selection, model design, and may expand existing social biases, leading to unfair treatment (Tsamados et al., 2021; Mitchell et al., 2020). Data bias begins from biased training data, algorithmic bias can happen even with impartial data if the model's structure introduces bias (Parikh et al., 2019). Concerns about algorithmic bias in ChatGPT arise from the lack of transparency regarding its training set and underlying language model (van Dis et al., 2023).
- *Data Privacy and Confidentiality:* Privacy issue is a major aspect when using ChatGPT in healthcare settings (Naik et al. 2022). Handling sensitive patient data, including collection, storage, and processing, raises privacy concerns regarding potential illegal entry or data leaks. ChatGPT's interactions with healthcare workers and patients entail gathering personal health information like medical histories and test results. Protecting this data is crucial for maintaining patient privacy and complying with laws like HIPAA (Su et al. 2021; US Department of Health and Human Services n.d.). Establishing a robust data governance framework ensures compliance with privacy regulations and promotes responsible data management.
- *Legal Responsibility:* The ability of ChatGPT to give incorrect medical counsel creates substantial legal implications (WAME 2023). AI currently lacks legal status, making humans the ultimate duty-bearers (Zhang and Zhang 2023). Determining legal responsibility when a patient is harmed can be complex, raising questions about accountability—whether it lies with the patient, hospital, or OpenAI. This highlights the necessity for clear legal contexts to define responsibility for AI use in healthcare. OpenAI has published security standards and guidelines, but they are not mandatory (OpenAI 2023a). ChatGPT's open access and OpenAI's disclaimer of responsibility for generated texts leave the burden of errors on users (OpenAI 2023b).

3.4 Justification of Identified Issues

- *Regulatory Compliance:* Integrating ChatGPT into medical chatbots requires adherence to regulations like HIPAA and GDPR to safeguard patient data and ensure compliance. Proactive regulatory frameworks are needed to address the dynamic nature of AI and protect patient rights.
- *Explainability and Transparency:* These are vital for developing confidence in AI systems. ChatGPT's transparency allows scrutiny of its decision-making, mitigating concerns about bias and errors, while explainability ensures comprehension of recommendations, fostering trust.

- *Algorithmic Bias*: Detecting and mitigating bias in AI algorithms is crucial to prevent disparities in healthcare delivery. Transparency in data usage and decision-making processes is necessary to effectively address biases and promote equitable healthcare outcomes.
- *Data Privacy and Confidentiality*: Guarding patient privacy is utmost in AI applications. Healthy security procedures and compliance with regulations like HIPAA are necessary to thwart unauthorized access and conserve patient trust throughout the AI lifecycle.
- *Legal Responsibility*: Clear legal frameworks are needed to define accountability for AI-generated outcomes. Establishing guidelines can help navigate legal challenges and allocate responsibility among developers, providers, and users, ensuring responsible AI use in healthcare (Wang et al., 2023).

4. Governance And Compliance Strategy

4.1 Impact of EU Regulations on Ethical and Trust Issues

The impact of EU regulations, including the General Data Protection Regulation (GDPR) and Artificial Intelligence Act (AIA), on ethical and trust issues surrounding the incorporation of ChatGPT, version 3–4 into medical chatbots is profound and multi-faceted.

- *Compliance Requirements and Legal Frameworks*: The AIA and GDPR establish comprehensive legal frameworks for AI and data protection within the European Union. The AIA aims to harmonize rules on artificial intelligence, including AI used in healthcare settings, by providing clear requirements for users and developers (European Parliament, 2024). Meanwhile, the GDPR places true criteria for the processing and securing of individual data, including healthcare data, which applies to medical chatbots utilizing ChatGPT (European Union, 2016). Adhering to these regulations is essential for healthcare organizations to ensure legal compliance and mitigate the risk of penalties.
- *Risk Assessment and Mitigation*: Both the AIA and GDPR emphasize the importance of risk assessment and mitigation tactics in the creation and implementation of current AI solutions. Medical chatbots leveraging ChatGPT may be considered high-risk under the AIA due to their potential impact on patient safety and health (European Parliament, 2024). Consequently, healthcare organizations must conduct thorough risk assessments, including assessments of algorithmic bias and data privacy risks, to guarantee the reliable and ethical application of ChatGPT in medical settings.
- *Transparency and Accountability*: Transparency and accountability are core principles embedded in both the AIA and GDPR. The AIA requires transparency regarding the use of AI systems, including explanations of their functions and decision-making processes (European Parliament, 2024). Similarly, the GDPR mandates transparency regarding data processing activities, securing individuals' awareness regarding how their data is handled, including health data processed by medical chatbots, is used (European Union, 2016). Healthcare organizations integrating ChatGPT into medical chatbots must prioritize transparency to cultivate trust amidst patients and medical practitioners.

- *Data Security and Privacy:* The GDPR has set strict requirements for the protection of private data, including health data processed by medical chatbots. Healthcare institutions ought to enforce healthy measures for securing data, such as log-on controls, and encryption, and to safeguard patient privacy and comply with GDPR requirements (European Union, 2016). Additionally, the AIA emphasizes the impact of data security and privacy in AI systems, further reinforcing the need for strict data privacy measures in medical chatbots utilizing ChatGPT (European Parliament, 2024).

4.2 Critical Evaluation of Governance and Compliance Measures

- *Regulatory Compliance and Governance Frameworks:* The incorporation of AI technologies like ChatGPT into healthcare delivery necessitates robust governance frameworks to ensure adherence to regulatory norms like the General Data Protection Regulation (GDPR) and the Health Insurance Portability and Accountability Act (HIPAA). While rules give cardinal guiding principles for data security and privacy, there are challenges in implementation and enforcement.
- *Algorithmic Bias Mitigation:* Dealing with algorithmic prejudice in AI systems like ChatGPT is essential to ensure equitable healthcare outcomes. While techniques such as bias detection algorithms and fairness-aware learning have been proposed to mitigate bias, their effectiveness remains limited. There is a significant bias in facial recognition systems, signaling the pervasive nature of algorithmic bias across various AI applications (Buolamwini and Gebu, 2018). To address this, healthcare organizations must adopt a holistic approach that involves diverse data collection, rigorous bias testing, and ongoing monitoring to detect and mitigate biases effectively.
- *Data Privacy and Confidentiality:* Guarding patient privacy and confidentiality is utmost when integrating AI technologies like ChatGPT into healthcare settings. While regulations like HIPAA and GDPR provide guidelines for data protection, there are ongoing concerns about data breaches and unauthorized access. A recent data breach at Norton Healthcare exposed sensitive patient information, emphasizing the persistent threat to data privacy in healthcare (Alder, 2023). To reduce these hazards, healthcare institutions need to give priority to cybersecurity precautions, including encryption, access controls, and regular security audits.
- *Legal Responsibility:* Determining legal responsibility for AI-generated outcomes remains a complex issue in healthcare. While developers like OpenAI provide security standards and guidelines, the allocation of legal responsibility in case of errors or harm is unclear. There is a need for clear legal frameworks to define responsibility for AI use in healthcare and protect users (Hui et al., 2021). Without such frameworks, there exists a possibility of legal uncertainty and potential accountability concerns for both developers and healthcare providers.

4.3 Development of Ethical Governance Framework for ChatGPT in Healthcare

The incorporation of ChatGPT into healthcare settings necessitates the development of a robust ethical governance framework to guarantee the accountable ethical usage of this AI technology.

This framework should address key ethical considerations, including transparency, accountability, fairness, privacy, and safety.

- ***Transparency:*** The governance framework should prioritize transparency to ensure that healthcare professionals and patients understand how ChatGPT operates, makes decisions, and generates responses. This includes providing information on the training processes, data sources, and management decision-making algorithms used by ChatGPT. Transparency promotes trust and facilitates users to assess the dependability and credibility of ChatGPT's recommendations.
- ***Fairness:*** Ensuring fairness in ChatGPT's interactions with patients is essential for mitigating algorithmic bias and promoting equitable healthcare outcomes. The governance framework should include measures to identify and reduce biases in training ChatGPT's algorithms and data. In addition, regular monitoring, and auditing of ChatGPT's performance can help detect and address any discriminatory patterns or disparities in its recommendations.
- ***Privacy:*** Guarding patient privacy and confidentiality is vital when deploying ChatGPT in healthcare settings. The governance framework should incorporate healthy data privacy measures, that include encryption, anonymization, and access control methods to secure sensitive health records. Conformity with the General Data Protection Regulation (GDPR) and Insurance Portability and Accountability Act (HIPAA) privacy regulations have to be ensured.
- ***Safety:*** The governance framework should prioritize patient safety by assessing and mitigating impending risks linked with ChatGPT's use in clinical practice. This includes evaluating ChatGPT's accuracy, reliability, and potential for harm in healthcare decision-making. Regular monitoring of ChatGPT's performance and user feedback can help identify and address safety concerns in real time.

5. Conclusion

In conclusion, integrating Chat Generative Pre-Trained Transformer (ChatGPT) into healthcare offers significant potential but also raises crucial governance, trust, and ethical concerns. Compliance with EU regulations like the AIA and GDPR is paramount, alongside transparency, bias mitigation, privacy protection, and defining legal responsibilities. Developing an ethical governance framework is essential to ensure responsible ChatGPT use, emphasizing transparency, accountability, fairness, privacy, and safety. Collaboration between stakeholders is vital to tackle these problems effectively and realize the values of AI-driven healthcare delivery alongside sustaining ethical standards and patient welfare.

References

1. Akter, S., Dwivedi, Y. K., Sajib, S., Biswas, K., Bandara, R. J., & Michael, K. (2022). Algorithmic bias in machine learning-based marketing models. *Journal of Business Research*, 144, 201-216. doi: 10.1016/j.jbusres.2022.02.013
2. Alder, S. (2023). Norton Healthcare Data Breach: Second Class Action Lawsuit Filed. *HIPAA Journal*. Retrieved from <https://www.hipaajournal.com/norton-healthcare-data-breach/>
3. Anderson, N., Belavy, D.L., Perle, S.M., et al. (2023). "AI did not write this manuscript, or did it? Can we trick the AI text detector into generated texts? The potential future of ChatGPT, and AI in Sports and Exercise Medicine manuscript generation." *BMJ Specialist J.*, 12, e001568. doi: 10.1136/bmjsem-2023-001568.
4. Balasubramaniam, N., Kauppinen, M., Rannisto, A., Hiekkanen, K., and Kujala, S. (2023). Transparency and explainability of AI systems: from ethical guidelines to requirements. *Information and Software Technology*, 159, 107197.
5. Barak-Corren, Y., Wolf, R., Rozenblum, R., Creedon, J.K., Lipsett, S.C., Lyons, T.W., Michelson, K.A., Miller, K.A., Shapiro, D.J., Reis, B.Y. and Fine, A.M. (2024). Harnessing the Power of Generative AI for Clinical Summaries: Perspectives From Emergency Physicians. *Annals of Emergency Medicine*. [online] Available at: <https://www.sciencedirect.com/science/article/pii/S0196064424000787> [Accessed 18 May 2024].
6. Brown, T. B., Mann, B., Ryder, N., Subbiah, M., Kaplan, J., Dhariwal, P., ... & Amodei, D. (2020). Language models are few-shot learners. In *Advances in Neural Information Processing Systems* 33.
7. Buolamwini, J., & Gebru, T. (2018). Gender shades: Intersectional accuracy disparities in commercial gender classification. *Proceedings of the 1st Conference on Fairness, Accountability and Transparency*, 77-91.
8. Caruana, R., Lou, Y., Gehrke, J., Koch, P., Sturm, M., & Elhadad, N. (2015). Intelligible models for healthcare: Predicting pneumonia risk and hospital 30-day readmission. *Proceedings of the 21th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 1721-1730.
9. Cascella, M.; Montomoli, J.; Bellini, V.; Bignami, E. (2023). Evaluating the feasibility of ChatGPT in healthcare: An analysis of multiple clinical and research scenarios. *J. Med. Syst.* 2023, 47, 33.
10. Chakraborty, C., Pal, S., Bhattacharya, M., Lee, S.-S. (2023). Overview of Chatbots with special emphasis on artificial intelligence-enabled ChatGPT in medical science. *Frontiers in Artificial Intelligence*, 6. DOI: 10.3389/frai.2023.1237704.
11. Corsello, A., and Santangelo, A. (2023). May artificial intelligence influence future pediatric research? —The Case of ChatGPT. *Children* 10, 757. doi: 10.3390/children10040757
12. Dave, T., Athaluri, S. A., and Singh, S. (2023). ChatGPT in medicine: an overview of its applications, advantages, limitations, future prospects, and ethical considerations. *Front. Artif. Int.* 6:1169595. doi: 10.3389/frai.2023.1169595
13. European Parliament. (2024). Proposal for a regulation of the European Parliament and of the Council laying down harmonised rules on artificial intelligence (Artificial Intelligence

- Act) and amending certain Union legislative acts - Outcome of the European Parliament's first reading (Strasbourg, 11 to 14 March 2024).
14. European Union. (2016). Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data, and repealing Directive 95/46/EC (General Data Protection Regulation).
 15. Guo, B., Zhang, X., Wang, Z., et al. (2023). "How close is ChatGPT to human experts? Comparison corpus, evaluation, and detection." doi: 10.48550/arXiv.2301.07597.
 16. Harrer, S. (2023). Attention is not all you need: The complicated case of ethically using large language models in healthcare and medicine. *eBioMedicine* 2023, 90, 104512.
 17. Hügler, T. (2023). "The wide range of opportunities for large language models such as ChatGPT in rheumatology." *RMD Open*, 9, e003105. doi: 10.1136/rmdopen-2023-003105.
 18. Hui, A. T., Ahn, S. S., Lye, C. T., & Deng, J. (2021). Ethical Challenges of Artificial Intelligence in Health Care: A Narrative Review. *Ethics in Biology, Engineering and Medicine: An International Journal*, 12(1), 55-71. Retrieved from <https://www.dl.begellhouse.com/journals/6ed509641f7324e6,0035cde100d29cae,3d9a82dd6d28fd1c.html>
 19. Jalil, S., Rafi, S., LaToza, T.D., et al. (2023). "ChatGPT and Software Testing Education: Promises and Perils." Piscataway, NJ: IEEE.
 20. Jang, M., and Lukasiewicz, T. (2023). "Consistency analysis of ChatGPT." doi: 10.48550/arXiv.2303.06273.
 21. King, M.R. (2023). "The future of AI in medicine: a perspective from a Chatbot." *Annal. Biomed. Eng.*, 51, pp.291–295. doi: 10.1007/s10439-022-03121-w.
 22. Kordzadeh, N., & Ghasemaghaei, M. (2022). Algorithmic bias: review, synthesis, and future research directions. *European Journal of Information Systems*. doi: 10.1080/0960085X.2022.204388
 23. Lan, X. and Zhenjing, L. (2022) 'The EU AI Act: A Risk-Based Approach to Regulating Artificial Intelligence', *Harvard Journal of Law & Technology*, 35(2), pp. 501-558.
 24. Lee, P., Bubeck, S., and Petro, J. (2023). "Benefits, limits, and risks of GPT-4 as an AI Chatbot for medicine." *New Engl. J. Med.*, 388, pp.1233–1239. doi: 10.1056/NEJMSr2214184.
 25. Liu, J., Wang, C., and Liu, S. (2023). Utility of ChatGPT in clinical practice. *J. Med. Int. Res.* 25:e48568. doi: 10.2196/48568
 26. Liu, S., Wright, A.P., Patterson, B.L., Wanderer, J.P., Turer, R.W., Nelson, S.D., McCoy, A.B., Sittig, D.F. and Wright, A. (2023). Using AI-generated suggestions from ChatGPT to optimize clinical decision support. *Journal of the American Medical Informatics Association*, 30(7), pp.1237-1245. <https://doi.org/10.1093/jamia/ocad072>.
 27. Liu, X., Glocker, B., McCradden, M.M., Ghassemi, M., Denniston, A.K. and Oakden-Rayner, L. (2022). The medical algorithmic audit. *Lancet Digit Health*, 4(5), pp.e384-e397. [FREE Full text] [online] Available at: [[https://doi.org/10.1016/S2589-7500\(22\)00010-5](https://doi.org/10.1016/S2589-7500(22)00010-5)] [Accessed 18 May 2024].
 28. Loh, E. (2023). "ChatGPT and generative AI chatbots: challenges and opportunities for science, medicine and medical leaders." *BMJ Leader*. doi: 10.1136/leader-2023-000797
 29. M. -N. Chu. (2023) "Assessing the Benefits of ChatGPT for Business: An Empirical Study on Organizational Performance," in *IEEE Access*, vol. 11, pp. 76427-76436, doi: 10.1109/ACCESS.2023.3297447.

30. Mitchell, M., Baker, D., Moorosi, N., Denton, E., Hutchinson, B., Hanna, A., et al. (2020). Diversity and inclusion metrics in subset selection. Presented at: AIES '20: AAAI/ACM Conference on AI, Ethics, and Society; Feb 7-9, 2020;117-123; New York, NY. doi: 10.1145/3375627.3375831
31. Mohd Javaid, Abid Haleem, Ravi Pratap Singh, (2023). "ChatGPT for healthcare services: An emerging stage for an innovative perspective", BenchCouncil Transactions on Benchmarks, Standards and Evaluations, vol. 3, no. 1, article 100105, ISSN 2772-4859, DOI: 10.1016/j.tbench.2023.100105.
32. Müller, V.C. (2012) 'Data for Humanity: An Open Letter', Ethics and Information Technology, 14(4), pp. 289-298.
33. Naik, N., Hameed, B. M. Z., Shetty, D. K., Swain, D., Shah, M., Paul, R., et al. (2022). Legal and ethical consideration in artificial intelligence in healthcare: who takes responsibility? Frontiers in Surgery, 9, 862322.
34. Naimi, L. (2023). Generative AI: An Introduction and Application with ChatGPT. Paper presented at The international conference on "Language and Artificial Intelligence: Celebratory Contexts and Investment Pathways", December 20 - 21, 2023, Errachidia, Morocco.
35. OpenAI. (n.d.). Privacy policy. Retrieved from <https://openai.com/en-GB/policies/privacy-policy> [Accessed 18 May 2024]
36. OpenAI. (2023). Introducing ChatGPT. [online] Available at: <https://openai.com/blog/chatgpt> [Accessed 18 May 2024].
37. OpenAI. 2023. Terms of use. [online] Available at: <https://openai.com/policies/terms-of-use> [Accessed 18 May 2024].
38. Ovalle, A., Beikzadeh, M., Teimouri, P., Chang, K., and Sarrafzadeh, M. (2023). ChatGPT for us: preserving data privacy in ChatGPT via dialogue text ambiguity to expand mental health care delivery. doi: 10.48550/arXiv.2306.05552
39. Parikh, R.B., Teeple, S. and Navathe, A.S. (2019). Addressing bias in artificial intelligence in health care. JAMA, [online] Available at: [<https://doi.org/10.1001/jama.2019.18058>] [Accessed 18 May 2024].
40. Partnership on AI. Safety critical AI. [online] Available at: <https://partnershiponai.org/program/safety-critical-ai/> [Accessed 18 May 2024].
41. Pennestrì, F., and Banfi, G. (2022). Artificial intelligence in laboratory medicine: fundamental ethical issues and normative key-points. Clin. Chem. Lab. Med. 60, 1867–1874. doi: 10.1515/cclm-2022-0096
42. PMI (Project Management Institute) (2022) AI Harms Directive: Establishing Liability Rules for AI Systems. Available at: <https://www.pmi.org/ai-harms-directive> (Accessed: 14 May 2024).
43. Reddy, S. (2024). Generative AI in healthcare: an implementation science informed translational path on application, integration and governance. Implementation Science, 19(1). <https://doi.org/10.1186/s13012-024-01357-9>
44. Ribeiro, M. T., Singh, S., & Guestrin, C. (2020). "Why should I trust you?": Explaining the predictions of any classifier. Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 1135-1144.
45. Roberts, V., Malone, K., Moore, P., et al. (2020). "Peer teaching medical students during a pandemic." Med. Educ., 25, p.1772014. doi: 10.1080/10872981.2020.1772014.

46. Roller, S., Dinan, E., Goyal, N., Ju, D., Williamson, M., Liu, Y., ... & Weston, J. (2020). Recipes for building an open domain chatbot. arXiv preprint arXiv:2004.13637.
47. Rosário, A. T. (2024). Generative AI and Generative Pre-Trained Transformer Applications: Challenges and Opportunities. In A. Travassos Rosário (Ed.), *Making Art With Generative AI Tools* (Chapter 3). doi:10.4018/979-8-3693-1950-5.ch003
48. Sallam, M. (2023). ChatGPT utility in healthcare education, research, and practice: Systematic review on the promising perspectives and valid concerns. *Healthcare* 2023, 11, 887.
49. Sanderson, K. (2023). "GPT-4 is here: what scientists think." *Nature*, 615, p.773. doi: 10.1038/d41586-023-00816-5.
50. Singhal, K.; Azizi, S.; Tu, T.; Mahdavi, S.S.; Wei, J.; Chung, H.W.; Scales, N.; Tanwani, A.; Cole-Lewis, H.; Pfohl, S. (2023). Large language models encode clinical knowledge. *Nature* 2023, 620, 172–180.
51. Su, Z., McDonnell, D., Bentley, B. L., He, J., Shi, F., Cheshmehzangi, A., et al. (2021). Addressing bio disaster X threats with artificial intelligence and 6G technologies: literature review and critical insights. *Journal of Medical Internet Research*, 23(5), e26109.
52. Tsamados, A., Aggarwal, N., Cows, J., Morley, J., Roberts, H., Taddeo, M., et al. (2021). The ethics of algorithms: key problems and solutions. *AI & Society*, 37(1), 215-230. doi: 10.1007/s00146-020-01154-8
53. US Department of Health and Human Services. (n.d.). Summary of the HIPAA security rule. Retrieved from <https://www.hhs.gov/hipaa/for-professionals/security/laws-regulations/index.html> [Accessed 18 May 2024]
54. Vaidyam, A. N., Wisniewski, H., Halamka, J. D., Kashavan, M. S., & Torous, J. B. (2019). Chatbots and Conversational Agents in Mental Health: A Review of the Psychiatric Landscape. *The Canadian Journal of Psychiatry*, 64(7), 456-464. doi:10.1177/0706743719828977
55. van Dis, E.A.M., Bollen, J., Zuidema, W., van Rooij, R. and Bockting, C.L. (2023). ChatGPT: five priorities for research. *Nature*, 614(7947), pp.224-226. [FREE Full text] [online] Available at: [<https://doi.org/10.1038/d41586-023-00340-6>] [Accessed 18 May 2024].
56. Wachter, S., Mittelstadt, B., and Floridi, L. (2017). Transparent, explainable, and accountable AI for robotics. *Sci Robot*, 2(6), ean6080.
57. WAME. (2023). Chatbots, generative AI, and scholarly manuscripts. [online] Available at: <https://wame.org/page3.php?id=106> [Accessed 18 May 2024].
58. Wang, C., Liu, S., Yang, H., Guo, J., Wu, Y., & Liu, J. (2023). Ethical considerations of using ChatGPT in health care. *Journal of Medical Internet Research*, 25, e48009. doi: 10.2196/48009.
59. Williams, R., Cloete, R., Cobbe, J., Cottrill, C., Edwards, P., Markovic, M., et al. (2022). *From transparency to accountability of intelligent systems: moving beyond aspirations*. Data & Policy, Cambridge University Press.
60. Winfield, A.F.T. and Jirotko, M. (2018) 'The case for an ethical black box', in *Towards Autonomous Robotic Systems*. Springer, pp. 262-273.
61. Zanca, F., Brusasco, C., Pesapane, F., Kwade, Z., Beckers, R., & Avanzo, M. (2022). Regulatory aspects of the use of artificial intelligence medical software. *Seminars in Radiation Oncology*, 32(4), 432-441.

62. Zhang, J., and Zhang, Z. (2023). Ethics and governance of trustworthy medical artificial intelligence. *BMC Medical Informatics and Decision Making*, 23(1), p.7. [online] Available at: <https://doi.org/10.1186/s12911-022-01813-8> [Accessed 18 May 2024].